CENTRO DE ENSEÑANZA TÉCNICA Y SUPERIOR

Escuela de Ingeniería

Machine Learning

Unit 1 - Homework 1

Teacher: Cesar Alan Torres Quiroz

Presented by:

Martin Alejandro Hermosillo Vargas - T028361

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1. **Female survivors older than 50**

Question: How many female passengers survived who were older than 50?

Answer: 16

Pandas Code:
```
print("1:", df[(df["Sex"] == "female") & (df["Survived"] == 1) &
(df["Age"] > 50)].shape[0])
```

2. **Median fare of passengers who did not survive**

Question: What was the median fare paid by passengers who did not survive?

Answer: 10.5

Pandas Code:
```
print("2:", df[df["Survived"] == 0]["Fare"].median())
```

3. **Third-class survival rate**

Question: What percentage of third-class passengers (Pclass == 3) survived?
Report the result as a percentage rounded to two decimal places.

Answer: 24.24

Pandas Code:
```
print("3:", round(df[df["Pclass"] == 3]["Survived"].mean() * 100, 2))
```

4. **Passengers by embarkation port**

Question: How many passengers embarked from each port (S, C, and Q)? How many rows have a missing Embarked value?

Answer:
S: 644
C: 168
Q: 77
Missing: 2

Pandas Code:
```
print("4 S:", (df["Embarked"] == "S").sum())
print("4 C:", (df["Embarked"] == "C").sum())
print("4 Q:", (df["Embarked"] == "Q").sum())
print("4 Missing:", df["Embarked"].isna().sum())
```

5. **Average age of survivors by sex**

Question: What was the average age of male survivors? What was the average age of female survivors? Which group was older on average?

Answer:
female : 28.847716
male : 27.276022
The group older on average is: female survivors

Pandas Code:
```
print("5:", df[df["Survived"] == 1].groupby("Sex")["Age"].mean())
```

6. **Missing cabin and survival status**

Question: How many passengers with no recorded Cabin did not survive?

Answer: 481

Pandas Code:
```
print("6:", df[df["Cabin"].isna() & (df["Survived"] == 0)].shape[0])
```

7. **Traveling alone and survival**

Question: Compare the survival rate of:
Passengers traveling completely alone: SibSp == 0 and Parch == 0
Passengers traveling with at least one family member aboard
Which group had the higher survival rate, and by how many percentage points?

Answer:
Alone: 30.35
Family: 50.56
Difference: 20.21

The group with the highest survival rate is: Passengers with family on board by 20.21 percentage points.

Pandas Code:

```
alone = df[(df["SibSp"] == 0) & (df["Parch"] == 0)]["Survived"].mean() *
100
family = df[(df["SibSp"] > 0) | (df["Parch"] > 0)]["Survived"].mean() *
100

print("7 Alone:", alone)
print("7 Family:", family)
print("7 Difference:", abs(alone - family))
```

8. **Average fare by passenger class**

Question: What was the average fare paid by passengers in each passenger class (Pclass 1, 2, and 3)?

Answer:
Pclass
1 84.154687
2 20.662183
3 13.675550

Pandas Code:

```
print("8:", df.groupby("Pclass")["Fare"].mean())
```

9. **Passengers younger than 18**

Question: How many passengers younger than 18 years old were aboard? What was their survival rate?

Answer:
Count: 113
Survival Rate: 53.98

Pandas Code:

```
young = df[df["Age"] < 18]
print("9 Count:", young.shape[0])
print("9 Survival rate:", young["Survived"].mean() * 100)
```

10. **Correlation between fare and passenger class**

Question: What is the correlation between Fare and Pclass?
State whether the correlation is positive or negative. In one sentence, explain whether the sign of the correlation matches your intuition and why.

Answer:
-0.55
The correlation between Fare and Pclass is approximately -0.55, which is a negative correlation. This sign matches intuition because lower class numbers represent higher socio-economic status (e.g., 1st class vs. 3rd class), meaning higher ticket prices correspond to lower numerical class values.

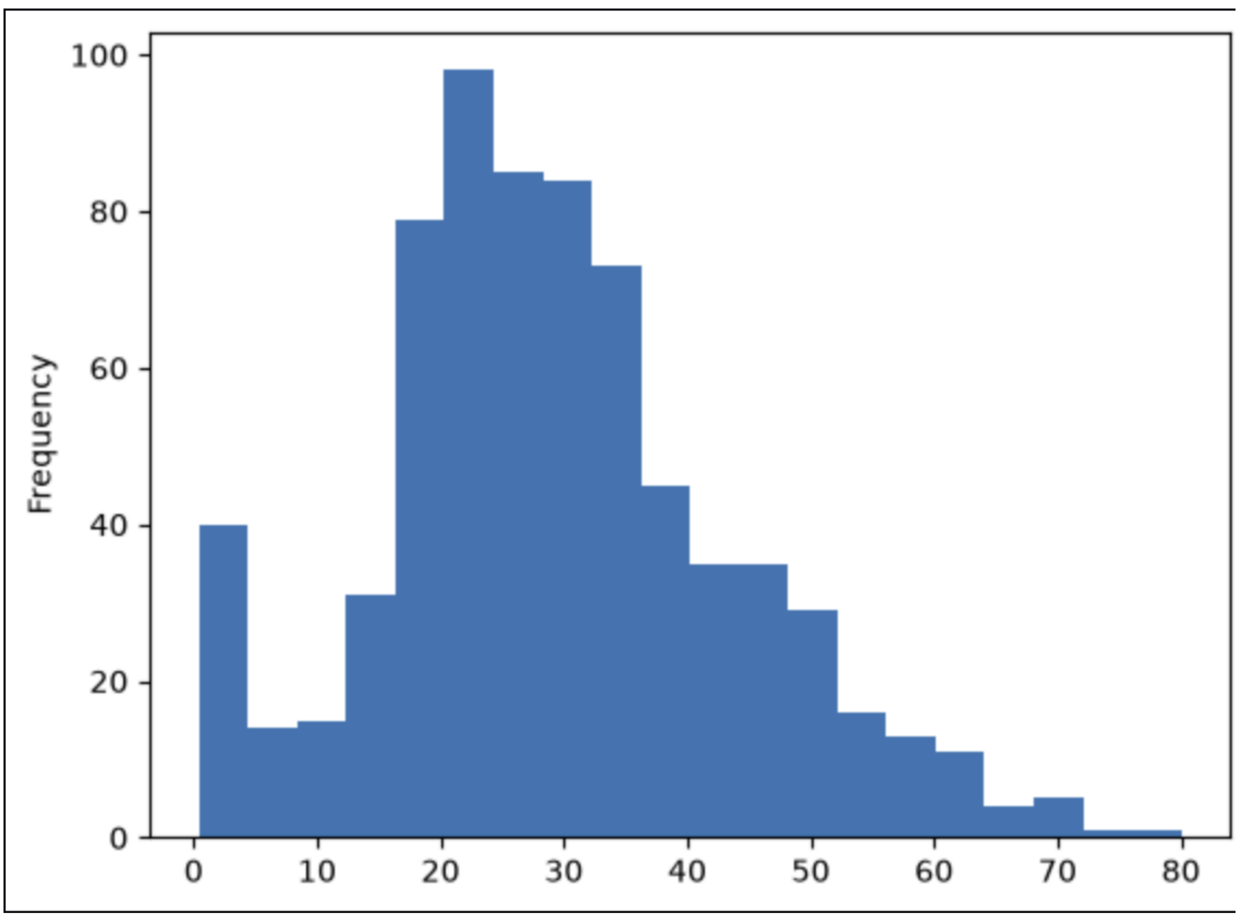
Pandas Code:

```
print("10:", df["Fare"].corr(df["Pclass"]))
```

11. **Age distribution plot**

Question: Create a histogram of the Age column.
Then answer: what does the distribution suggest about the age of the passengers, and where is the largest concentration of values?

Answer:



The distribution is right-skewed, showing that the majority of passengers were young to middle-aged adults, along with a notable peak for young children (under 5 years old). The largest concentration of values is located between 20 and 30 years of age.

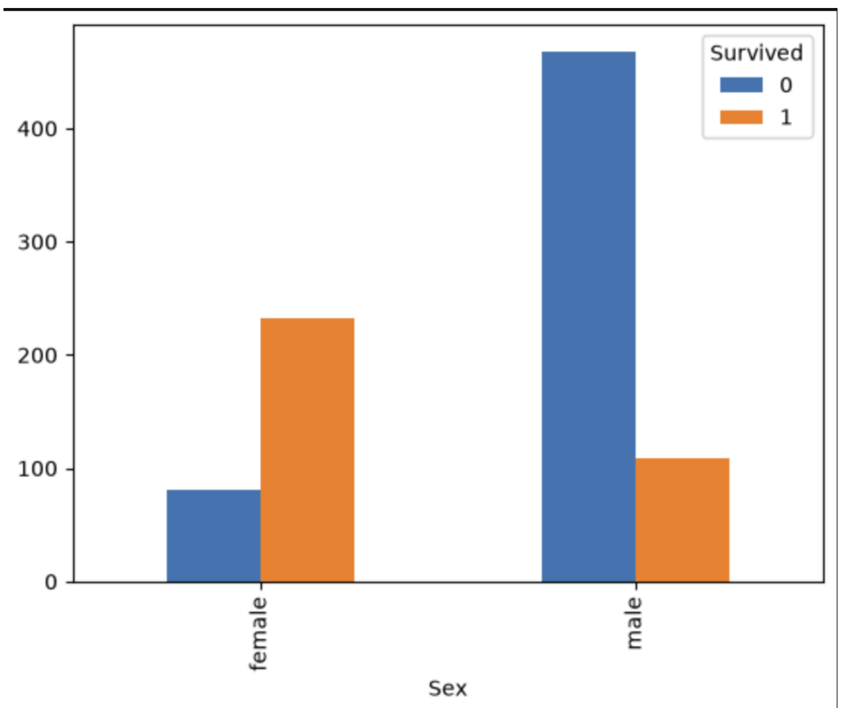
Pandas Code:

```
import matplotlib.pyplot as plt
df["Age"].plot.hist(bins=20)
plt.show()
```

12. **Survival by sex or passenger class**

Question: Create a bar chart showing the number of passengers who survived and did not survive by Sex or by Pclass.
Which group had the highest number of survivors, and what does the chart suggest about the relationship between the selected feature and survival?

Answer:



Female passengers had the highest number of survivors (over 230). The chart suggests a strong relationship between sex and survival, indicating that females were significantly more likely to survive than males, while the majority of male passengers did not survive.

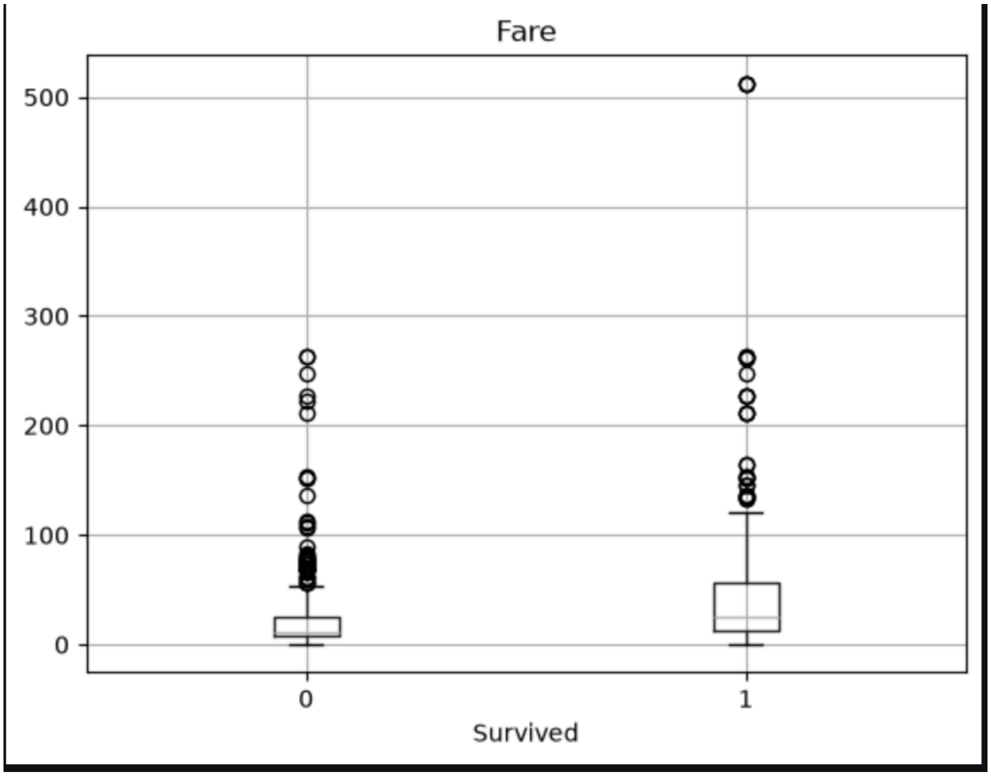
Pandas Code:

```
df.groupby(["Sex", "Survived"]).size().unstack().plot(kind="bar")
plt.show()
```

13. **Fare distribution by survival status**

Question: Create a box plot for Fare grouped by Survived.
What does the chart suggest about the typical fare paid by survivors versus non-survivors, and are there clear outliers?

Answer:



The chart suggests that survivors generally paid a higher typical fare than non-survivors, as indicated by the higher median line and larger interquartile range for the survived group (1). There are clear outliers in both categories, most notably extreme high-fare outliers among survivors, including a few tickets priced over 500.

Pandas Code:

```
df.boxplot(column="Fare", by="Survived")
plt.suptitle("")
plt.show()
```